

CHAPTER 2

How Search Engines Work

Learning Objectives

After completing this chapter, you will be able to:

- Understand how Google finds websites.
 - Learn what Crawling, Indexing, and Ranking mean.
 - Understand the role of Googlebot.
 - Learn how Google stores billions of web pages.
 - Understand why some websites rank first while others don't.
 - Learn what happens when you search on Google.
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Introduction

Imagine there are **more than 1.5 billion websites** on the Internet.

When you search for:

"Best AI Laptop Under ₹60,000"

Google gives you results in **less than one second**.

Have you ever wondered:

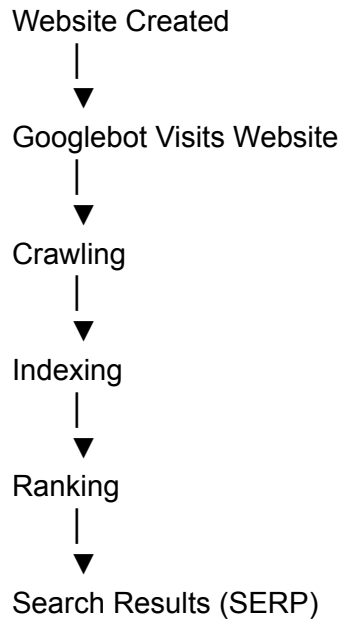
- How does Google know these websites exist?
- How does Google find the correct websites?
- Why is one website ranked first and another on page 10?

The answer lies in **three major processes**:

1. Crawling
2. Indexing
3. Ranking

These three steps are the foundation of every search engine.

The Complete Workflow



This is the journey every website goes through before it appears in Google Search.

Topic 1: What is Crawling?

Definition

Crawling is the process by which search engines discover new and updated web pages on the Internet.

Google uses special software called **Googlebot** (also known as a crawler or spider) to visit websites and collect information.

Think of crawling as a librarian walking through a library, checking every new book that has arrived.

Why Crawling is Needed

Google cannot rank a website unless it first knows that the website exists.

If your website is never crawled:

- Google does not know about it.

- It cannot be indexed.
- It cannot appear in search results.

Crawling is the **first step** toward visibility.

How Crawling Works

Suppose you create a new blog:

www.ananyatech.com

Googlebot discovers it through one or more of these methods:

- Following links from other websites.
- Reading your XML Sitemap.
- Discovering it from previous crawls.
- When you manually request indexing in Google Search Console.

Once discovered, Googlebot visits the pages and reads:

- Text
 - Images
 - Headings
 - Links
 - Metadata
 - Structured data
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What Does Googlebot Do?

Googlebot acts like an automated visitor.

It checks:

- What the page is about.
- Whether it loads properly.
- Whether links work.
- Whether the page is mobile-friendly.
- Whether new content has been added.

Googlebot keeps revisiting websites because content changes over time.

Real-Life Example

Imagine a teacher checks every classroom notice board every morning.

If a new notice is posted, the teacher reads it and notes the update.

Similarly, Googlebot periodically checks websites for new or updated content.

Topic 2: What is Googlebot?

Definition

Googlebot is Google's web crawler.

Its job is to continuously explore the web and collect information from websites.

Googlebot:

- Visits pages.
 - Reads content.
 - Follows links.
 - Discovers new pages.
 - Sends collected information back to Google's servers.
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Types of Googlebot

Google uses different crawlers for different purposes.

Common examples:

- Googlebot Desktop
 - Googlebot Smartphone (most important today because Google uses **mobile-first indexing**)
 - Googlebot Images
 - Googlebot Videos
 - Googlebot News
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Topic 3: What is Indexing?

Definition

After crawling a page, Google analyzes it and stores useful information in its **index**.

The index is like a giant digital library.

Instead of storing books, it stores information about billions of web pages.

Only indexed pages are eligible to appear in search results.

How Indexing Works

1. Googlebot crawls the page.
2. Google analyzes the content.
3. It understands the topic.
4. It stores important details in its index.

Information stored includes:

- Page title
 - Keywords and topics
 - Images
 - Videos
 - Internal links
 - External links
 - Metadata
 - Language
 - Structured data
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Example

Suppose your page is titled:

"Top 10 AI Projects for Engineering Students"

Google may index it under topics such as:

- AI Projects
- Engineering Projects
- Machine Learning
- Student Projects

Later, when someone searches for these topics, your page may be considered.

Can Every Page Be Indexed?

No.

Google may choose **not** to index pages if they:

- Have very little useful content.
 - Contain duplicate content.
 - Are blocked by `robots.txt`.
 - Use a `noindex` tag.
 - Return server errors.
 - Are inaccessible.
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Topic 4: What is Ranking?

Definition

Ranking is the process of deciding the order in which indexed pages appear in search results.

Google wants to show the **most relevant and useful** pages first.

How Google Decides Rankings

Google evaluates **hundreds of ranking signals**. Some important ones include:

- Content quality
- Relevance to the search query
- Page speed
- Mobile friendliness
- Secure connection (HTTPS)
- Quality of backlinks
- User experience
- Freshness of content
- Website authority
- Internal linking
- Search intent match

Google combines these signals to estimate which page best answers the user's question.

Example

Suppose three websites publish an article on:

"Best Budget Smartphones"

If:

- Website A has detailed, updated content, fast loading speed, and trusted backlinks,
- Website B has short, outdated content,
- Website C loads slowly and has broken links,

Google is more likely to rank Website A higher because it provides a better experience.

Topic 5: What is SERP?

Definition

SERP stands for **Search Engine Results Page**.

It is the page displayed after you perform a search.

A SERP may include:

- Organic search results
 - Paid advertisements
 - Featured snippets
 - Images
 - Videos
 - Maps
 - People Also Ask
 - Shopping results
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Example

Search:

"Python Tutorial"

The SERP might show:

1. Featured snippet

2. Video results
3. Organic websites
4. "People Also Ask" questions
5. Related searches

SEO aims to help your website appear prominently in these organic results.

Real-World Analogy

Think of Google as a **massive library**:

- **Googlebot** is the librarian collecting new books.
 - **Crawling** is finding the books.
 - **Indexing** is cataloging them.
 - **Ranking** is deciding which books to place on the front shelf for readers.
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Chapter Summary

- **Crawling**: Googlebot discovers web pages.
 - **Googlebot**: Google's automated crawler.
 - **Indexing**: Useful pages are stored in Google's index.
 - **Ranking**: Google orders pages based on relevance and quality.
 - **SERP**: The page where search results are displayed.
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Interview Questions

1. What is a search engine?
 2. What is Googlebot?
 3. What is crawling?
 4. What is indexing?
 5. What is ranking?
 6. Can a page rank if it is not indexed?
 7. What is the difference between crawling and indexing?
 8. What is SERP?
 9. Name some factors that influence ranking.
 10. Why is Googlebot important?
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Quick Revision Table

Term	Meaning
Crawler (Googlebot)	Software that discovers web pages
Crawling	Finding and reading web pages
Indexing	Storing information about pages in Google's index
Ranking	Deciding the order of search results
SERP	Search Engine Results Page shown after a search